

JASPER, AB, Canada

WMO: 728880

Lat: 52.880N

Lon: 118.070W

Elev: 3485

StdP: 12.94

Time Zone: -7.00 (NAM)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-24.1	-17.5	-28.7	1.3	-22.9	-22.8	1.8	-17.1	17.0	7.5	15.5	9.7	4.7	230	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	81.3	57.9	77.8	56.7	74.4	55.3	59.6	76.4	58.1	73.9	56.6	71.1	5.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	71.1	61.3	52.3	66.2	59.6	50.5	62.0	59.2	27.7	76.7	26.6	74.1	25.5	71.1	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.3	11.6	DB	-31.3	86.8	8.5	1.7	-37.4	88.1	-42.3	89.0	-47.1	90.0	-53.3	91.2	(4)
(5)			WB	-30.5	63.2	7.6	1.0	-36.0	64.0	-40.4	64.6	-44.7	65.2	-50.2	65.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	18.0	22.7	30.9	39.6	48.0	55.5	59.0	57.9	49.2	39.7	24.1	16.1	
	DBStd	18.25	14.79	15.25	9.08	7.12	6.22	5.19	5.36	5.92	7.43	10.10	13.13	14.96	
	HDD50	5069	992	766	592	316	116	13	2	6	104	334	776	1052	
	HDD65	9703	1457	1186	1057	761	528	287	198	228	475	783	1226	1517	
	CDD50	863	0	0	0	4	54	177	281	252	79	16	0	0	
	CDD65	23	0	0	0	0	0	1	12	9	1	0	0	0	
	CDH74	779	0	0	0	1	49	81	343	254	50	1	0	0	
	CDH80	135	0	0	0	0	9	6	74	37	9	0	0	0	
Wind	WSAvg	4.5	4.9	4.9	4.3	4.9	4.7	4.5	4.1	3.8	3.8	4.6	4.5	4.9	
Precipitation	PrecAvg	16.3	1.3	0.7	0.7	0.9	1.3	2.1	2.2	2.1	1.6	1.3	1.1	1.1	
	PrecMax	21.2	3.9	2.4	1.9	1.9	3.8	4.3	5.4	5.1	4.3	2.8	2.4	3.4	
	PrecMin	12.9	0.1	0.0	0.1	0.1	0.4	0.7	0.3	0.7	0.6	0.3	0.2	0.2	
	PrecStd	2.4	0.9	0.6	0.4	0.5	0.7	1.0	1.3	1.1	0.7	0.6	0.6	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	49.6	56.5	69.6	81.2	81.5	85.7	83.9	79.8	71.1	47.6	43.8	
		MCWB	34.7	39.0	41.1	50.1	56.1	56.2	59.2	59.9	57.6	53.3	39.9	35.1	
	2%	DB	38.7	44.9	50.5	62.8	73.0	76.7	82.0	80.2	73.6	63.1	43.9	39.1	
		MCWB	34.0	36.8	37.7	45.8	51.8	54.8	58.4	58.5	54.6	49.4	38.5	33.9	
	5%	DB	36.7	41.8	46.6	58.5	67.9	73.1	77.9	76.7	68.3	58.3	41.0	35.9	
		MCWB	33.0	35.1	36.3	43.5	49.8	53.9	56.7	57.4	53.0	46.8	36.6	32.3	
	10%	DB	34.4	39.1	43.2	53.4	62.8	69.0	73.6	72.7	62.9	53.2	37.7	33.4	
		MCWB	31.5	34.0	34.8	41.0	47.2	51.7	55.4	55.8	50.5	43.9	34.2	30.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	40.7	42.2	51.3	56.5	58.2	61.1	61.8	58.3	53.7	41.2	36.5
			MCDB	39.4	48.5	51.9	67.0	78.6	73.7	79.7	77.3	77.3	70.0	46.2	41.6
2%		WB	34.8	37.8	39.4	46.6	52.8	56.5	59.4	60.1	55.5	50.2	39.3	34.4	
		MCDB	37.4	43.1	48.1	60.6	70.0	72.5	77.5	75.7	71.0	62.4	43.7	38.4	
5%		WB	33.1	35.8	37.4	44.2	50.5	54.9	57.8	58.6	53.4	47.3	36.9	32.5	
		MCDB	36.3	40.4	44.7	55.8	65.6	69.8	74.2	73.5	66.3	56.8	40.1	36.1	
10%		WB	31.4	34.2	35.4	42.1	48.2	53.4	56.3	57.0	51.6	44.7	34.6	30.6	
		MCDB	34.6	39.0	42.6	52.5	61.2	66.3	70.9	69.8	62.0	51.8	37.6	33.3	
Mean Daily Temperature Range	5% DB	MDBR	14.5	17.2	19.6	21.4	22.0	21.7	22.8	23.1	22.3	18.4	14.0	13.5	
		MCDBR	12.4	17.0	23.7	31.8	33.1	31.1	33.2	31.4	33.5	27.4	14.1	11.3	
		MCWBR	9.0	10.8	13.6	17.0	15.7	13.6	13.4	13.3	16.5	15.2	9.4	7.9	
	5% WB	MCDBR	12.2	15.8	21.4	27.7	28.7	25.9	29.0	28.6	29.8	25.5	13.5	11.1	
		MCWBR	9.4	10.4	12.9	15.2	14.2	11.8	12.4	12.3	15.1	14.4	9.4	8.1	
Clear-Sky Solar Irradiance	taub	0.256	0.263	0.298	0.347	0.364	0.388	0.365	0.366	0.336	0.298	0.262	0.251		
	taud	2.320	2.363	2.271	2.155	2.256	2.270	2.386	2.374	2.449	2.507	2.390	2.248		
	Ebn at Noon	238	275	283	280	280	274	279	272	268	259	234	213		
	Edh at Noon	18	24	33	42	40	40	35	34	28	21	17	16		
All-Sky Solar Radiation	RadAvg	255	526	852	1316	1575	1591	1691	1453	1002	557	267	200		
	RadStd	25	56	86	125	161	139	149	89	117	62	32	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page